

GRIPABLE



GripAble Sensor: Research Evidence

V4 JANUARY 2026





GripAble is an evidence-based sensor for functional assessment and rehabilitation

Grip strength, a key biomarker for overall health and functional capacity, is pivotal in predicting chronic diseases and individual's future health risk to frailty, disability, falls, and mortality [1, 2]. GripAble is a validated digital dynamometer offering accurate and sensitive grip strength assessment [3], with an excellent test-retest and interrater reliability [4, 5]. It has recently established normative grip strength datasets for UK [5] and US [publication pending] populations. GripAble aspires to build the world's largest grip strength dataset by gathering data from users worldwide, enabling users to compare their strength results with those of their peers, including individuals with similar demographics or health conditions.

The Able Assess app incorporates digitised mobility assessments using the GripAble sensor to provide a comprehensive falls screening tool. The Timed-Up-and-Go (TUG) test, 30-sec Chair Stand Test (CST) and 4-metre Gait Speed (4GS) test alongside grip strength assessment have been selected based on internationally recognised guidelines and provide a multifactorial, objective and standardised view of falls risk [6, 7]. Building on existing validation evidence, further reliability studies including inter-instrument, inter-rater and test-retest reliability of mobility assessments will be conducted. This will act as the basis for retrospective and prospective studies to validate predictive performance of falls history and future falls, respectively.

In rehabilitation, higher exercise repetition, duration and intensity is a key factor in achieving better outcomes [8], and gamification is an effective strategy for increasing engagement promoting higher intensity exercise [9]. Evidence shows that GripAble game-based rehabilitation activities yields remarkable effectiveness at doubling exercise duration and achieving eight times the number of repetitions compared to conventional care. This leads to clinically significant improvement in arm impairment and arm function [10] while demonstrating effective usability and acceptability across neurological and musculoskeletal conditions such as Parkinson's disease, cerebral palsy and fractures.

This document is a repository of research evidence supporting the scientific foundation of GripAble

1. Sayer AA, Syddall HE, Martin HJ, Dennison EM, Roberts HC, Cooper C. Is grip strength associated with health-related quality of life? Findings from the Hertfordshire Cohort Study. Age and ageing. 2006 Jul 1;35(4):409-15. [\[LINK\]](#)
2. Rantanen T, Guralnik JM, Foley D et al. Midlife hand grip strength as a predictor of old age disability. JAMA 1999; 281: 558-60. [\[LINK\]](#)
3. Mace M, Motalib SA, Ogrinc M, Goldsmith N, Burdet E. GripAble: An accurate, sensitive and robust digital device for measuring grip strength. Journal of Rehabilitation and Assistive Technologies Engineering. 2022 Feb 8;9:20556683221078455. [\[LINK\]](#)
4. Ergen HI, Kudin R, McGee CW. Interrater Reliability and Precision of a Novel Hand Strength Assessment and Treatment Device: The GripAble. The American Journal of Occupational Therapy. 2024 Sep 1;78(5) [\[LINK\]](#)
5. Motalib, Sharah Abdul, et al. "GripAble: Interrater reliability and normative grip strength of UK population." *Journal of Hand Therapy* (2024). [\[LINK\]](#)
6. Sarmiento K, Lee R. STEADI: CDC's approach to make older adult fall prevention part of every primary care practice. J Safety Res. 2017;63:105-9. [\[LINK\]](#)
7. Montero-Odasso M et al., World guidelines for falls prevention and management for older adults: a global initiative, Age and Ageing 2022; 51: 1-36 [\[LINK\]](#)
8. Lohse KR, Lang CE, Boyd LA. Is more better? Using metadata to explore dose-response relationships in stroke rehabilitation. Stroke. 2014 Jul;45(7):2053-8 [\[LINK\]](#)
9. Dumas I, Everard G, Dehem S, Lejeune T. Serious games for upper limb rehabilitation after stroke: a meta-analysis. Journal of neuroengineering and rehabilitation. 2021 Dec;18:1-6 [\[LINK\]](#)
10. Broderick M, Almedom L, Burdet E, Burridge J, Bentley P. Self-directed exergaming for stroke upper limb impairment increases exercise dose compared to standard care. Neurorehabilitation and Neural Repair. 2021 Nov;35(11):974-85. [\[LINK\]](#)



RESEARCH: SUMMARY

GRIPABLE FOR ASSESSING GRIP STRENGTH

9 studies validating GripAble
as a dynamometer

Validated **accuracy, sensitivity and robustness**, compared to gold-standard Jamar, with excellent reliability for measurements across raters and real world use cases.

Established UK normative dataset of healthy adults aged 18 to 90 years (N=907)

Ongoing collaboration with world's leading institution to publish US normative dataset (N=875), superseding 40-year-old outdated dataset



GRIPABLE FOR SCREENING FALLS RISK

1 study validating Able Assess
as a falls screening tool

Internal validation study demonstrates **close inter-instrument agreement** across three mobility assessments (N=117)

Retrospective analysis highlights **predictive potential of falls risk metric** with participant falls history

Establishing new research partnerships to accelerate validation of mobility assessments (i.e. inter-rater, test-retest, norms) and falls risk prediction



GRIPABLE FOR GAMIFIED REHABILITATION

20 publications across various
institutions and providers

Evidence for **clinically significant improvements in arm impairment and arm function**, as well as improvements in overall functional performance

Established evidence for **doubling exercise duration and 8x reps compared to standard care**. 87% usability with stroke patients; equally excellent usability with other neuro conditions e.g. Parkinson's, CP

Externally-led research demonstrating GripAble as a rehab tool for neurological, neurodegenerative and musculoskeletal conditions

IMPERIAL



GRIPABLE SENSOR FOR ASSESSING GRIP STRENGTH



GRIPABLE FOR ASSESSING GRIP STRENGTH: SUMMARY

Grip strength is now recognised as a biomarker for overall health and functional status [1]. It serves as an indicator of musculoskeletal and neuromuscular health and has applications in numerous fields including geriatrics, sports and rehabilitation medicine.

Grip Strength as a Biomarker:

1. **Health Assessment:** Grip strength is a reliable indicator of an individual's overall health and vitality. It correlates with various health parameters, including muscle mass, bone density, and cardiovascular health [1, 2, 3]
2. **Predictive Value:** Low grip strength is associated with an increased risk of various health conditions, such as frailty, disability, and mortality [4, 5]. It can predict the onset of chronic diseases like diabetes [6] and cardiovascular issues [2]. Routine grip strength measurement is recommended for identifying older adults at risk of poor health status [3].
3. **Cognitive and Functional Assessment:** Beyond health, grip strength is a practical measure of cognitive functions [7], ability to perform daily activities, such as lifting, carrying, and grasping objects, while assessing an individual's quality of life, functional capacity and independence [3, 4].

GripAble: A Validated Dynamometer for Measuring Grip Strength:

1. **Accurate, Sensitive and Robust:** GripAble is a robust state-of-the-art dynamometer designed specifically for measuring grip strength accurately and reliably. It employs advanced sensor technology to provide accurate, sensitive and repeatable measurements [8] (see Slide 6).
2. **Validated Dynamometer:** GripAble has undergone rigorous industry-standard validation studies, demonstrating its consistency in measuring grip strength across diverse populations to ensure that it can be used confidently in clinical and research settings [8 - 10] (see Slide 6).
3. **Normative Data:** GripAble has established UK [10] and US [publication pending] normative grip strength datasets.

1. Vaishya R et al. Hand grip strength as a proposed new vital sign of health: a narrative review of evidences. J. of Health, Population and Nutrition. 2024 43:7. [\[LINK\]](#)
2. Leong DP et al. Prognostic value of grip strength: findings from the Prospective Urban Rural Epidemiology (PURE) study. The Lancet. 2015 Jul 18;386(9990):266-73. [\[LINK\]](#)
3. Bohannon RW. Grip strength: an indispensable biomarker for older adults. Clinical interventions in aging. 2019 Oct 1:1681-91. [\[LINK\]](#)
4. Sayer AA et al. Is grip strength associated with health-related quality of life? Findings from the Hertfordshire Cohort Study. Age and ageing. 2006 Jul 1;35(4):409-15. [\[LINK\]](#)
5. Rantanen T et al. Midlife hand grip strength as a predictor of old age disability. JAMA 1999; 281: 558-60. [\[LINK\]](#)
6. Boonpor J et al. Associations between grip strength and incident type 2 diabetes: findings from the UK Biobank prospective cohort study BMJ Open Diabetes Research and Care 2021;9:e001865. [\[LINK\]](#)
7. Jiang R et al. Associations between grip strength, brain structure, and mental health in> 40,000 participants from the UK Biobank. BMC medicine. 2022 Dec;20(1):1-4. [\[LINK\]](#)
8. Mace M et al.. GripAble: An accurate, sensitive and robust digital device for measuring grip strength. Journal of Rehabilitation and Assistive Technologies Engineering. 2022 Feb 8;9:20556683221078455. [\[LINK\]](#)
9. Mutalib SA et al. Modernising grip dynamometry: Inter-instrument reliability between GripAble and Jamar. BMC Musculoskelet Disord. 2022 Jan 24;23(1):80. [\[LINK\]](#)
10. Mutalib SA et al. GripAble: Interrater reliability and normative grip strength of UK population. Journal of Hand Therapy. 2024 Mar 22. [\[LINK\]](#)



GRIPABLE FOR ASSESSING GRIP STRENGTH: INTERNAL VALIDATION STUDIES

Lead Institution	Title	Data/Evidence	Status
 GRIPABLE In collaboration with IMPERIAL	[1] GripAble: An accurate, sensitive and robust digital device for measuring grip strength (2022), Journal of Rehabilitation and Assistive Technologies Engineering <i>Michael Mace, Sharah Abdul Mutalib, Matjaz Ogrinc, Nicola Goldsmith, Etienne Burdet</i>	Bench testing evidence demonstrating GripAble's outstanding accuracy, sensitivity and robustness	Completed and published
	[2] Modernising grip dynamometry: Inter-instrument reliability between GripAble and Jamar (2022), BMC Musculoskeletal Disorders <i>Sharah Abdul Mutalib, Michael Mace, Chloe Seager, Etienne Burdet, Virgil Mathiowetz, Nicola Goldsmith</i>	Excellent agreement between GripAble and the gold-standard dynamometer, Jamar.	Completed and published
	[3] GripAble: Interrater reliability and normative grip strength of UK population (2024), Journal of Hand Therapy <i>Sharah Abdul Mutalib, Deepika Sharma, Sonia Pike, Samantha Hyde, Liz Gwynne, Jenn Morehouse, Helen Davey, Lauren Edwards, Pedro Douglass-Kirk, Etienne Burdet, Nicola Goldsmith, Michael Mace</i>	1 - Excellent interrater reliability, indicating generalisability and reliability for grip strength measurement by any raters. 2 - UK's first grip strength normative data using GripAble - 907 healthy adults aged 18 to 94 years	Completed and published
	[4] Remote, Reliable, Repeatable: Real-World Test-Retest Validation of Hand Grip Strength Assessments (pending publication) <i>Sharah Abdul Mutalib, Daniel Jenkinson, Sonia Pike, Etienne Burdet, Corey McGee, Michael Mace</i>	Good-to-Excellent test-retest reliability across 3 real-world scenarios (i.e. face-to-face supervision, video call supervision, independent unsupervised)	Data collection completed, pending publication
	[5] Hand Grip Strength as a Remote Health-Monitoring Metric: The Importance of User Compliance (internal report) <i>Daniel Jenkinson, Sharah Abdul Mutalib, Michael Mace</i>	1 - Investigates the effect of non-compliance to ASHT guidelines 2 - Development and validation of a machine learning model for remote compliance monitoring	Internal report



GRIPABLE FOR ASSESSING GRIP STRENGTH: EXTERNAL VALIDATION STUDIES

Lead Institution	Title	Data/Evidence	Status
 UNIVERSITY OF MINNESOTA	[6] Interrater Reliability and Precision of a Novel Hand Strength Assessment and Treatment Device: The GripAble (2024), The American Journal of Occupational Therapy <i>Halil Ibrahim Ergen, Roman Kudin, Corey W. McGee</i>	Independent study showing excellent interrater reliability and acceptable precision of GripAble at measuring grip strength	Completed and published
	[7] Test-Retest Reliability and Precision of GripAble, A Multi-Purpose Exergaming (2025), Games for Health Journal <i>Roman Kudin, Halil Ibrahim Ergen, Corey W. McGee</i>	Independent study showing excellent test-retest reliability and acceptable precision of GripAble at measuring grip strength	Completed and published
	[8] Normative grip strength of US population (pending publication) <i>Corey W. McGee</i>	US's first grip strength normative data using GripAble - >800 healthy adults aged 18 to 94 years	Data collection completed, pending publication
 MONASH University	[9] Establishing the reliability and validity for GripAble: A handheld sensor-based dynamometry device (2025), Journal of Hand Therapy <i>Elise Yee Yan Li, Luke Steven Robinson, Claire Stockdale</i>	Independent study showing excellent test-retest reliability and good-to-excellent inter-instrument reliability of GripAble at measuring grip strength	Completed and published



GRIPABLE FOR SCREENING FALLS RISK



GRIPABLE FOR SCREENING FALLS RISK: SUMMARY

An individual's risk of falling is a complex issue, influenced by multiple factors such as muscle strength, balance, mobility, and overall physical resilience [1, 2]. There are multiple international guidelines that provide frameworks for screening for falls risk including the Centers for Disease Control and Prevention Stopping Elderly Accidents, Death and Injuries (CDC STEADI) [3], the World Guidelines for Falls Prevention (WGFP) [4, 5], and NICE [6]. Able Assess incorporates 3 mobility assessments alongside grip strength to provide a comprehensive screening for falls risk stratification based on its multifactorial components.

Lower Limb Mobility Assessments:

1. **30-Second Chair Stand Test (CST):** A validated measure of lower body strength and endurance which is critical for mobility, independence, and fall prevention. The CDC STEADI recommends to use age and sex-specific cut-offs derived from a large US study of 7,183 community-dwelling older adults [7].
2. **Timed Up and Go (TUG) test:** This test is one of the most widely used and validated measures of dynamic balance, gait, and mobility with the CDC recommending a fixed cut-off of 12 seconds [8] despite age and sex-specific cut-offs being available [9].
3. **4-Meter Gait Speed (4GS) test:** A core measure of falls risk and functional decline recommended by multiple guidelines and considered a functional vital sign [10]. Cut-offs vary by population with the CDC STEADI recommending 0.8 m/s although recent studies also indicate more complex relationships between walking speed and falls [11].

Inter-Instrument Reliability Internal Study (N = 117) [12]:

1. **Close agreement with CDC measurements** based on Inter-Class Correlation ($ICC_{2,1}$) analysis
2. Able Assess falls risk shows **promising predictive performance** for self-reported falls history
3. **Grip Strength is a significant predictor** of recent falls

1. Montero-Odasso M et al., Evaluation of Clinical Practice Guidelines on Fall Prevention and Management for Older Adults: A Systematic Review. JAMA Network Open. 2021 [\[LINK\]](#)
2. Colón-Emeric CS, McDermott CL, Lee DS et al. Assessment and prevention of falls in older community-dwelling adults: a review. JAMA. 2024;331:1397–406. [\[LINK\]](#)
3. Sarmiento K, Lee R. STEADI: CDC's approach to make older adult fall prevention part of every primary care practice. J Safety Res. 2017;63:105–9. [\[LINK\]](#)
4. Montero-Odasso M et al., World guidelines for falls prevention and management for older adults: a global initiative, Age and Ageing 2022; 51: 1–36 [\[LINK\]](#)
5. Montero-Odasso M, Van der Velde N, Ryg J, Masud, T, Falls risk stratification. One size doesn't fit all, Age and Aging 2025;54 [\[LINK\]](#)
6. Falls: assessment and prevention in older people and in people 50 and over at higher risk. NICE guidelines (NG249 - April 2025) [\[LINK\]](#)
7. Rikli R, Jones C, Functional fitness normative scores for community-residing older adults, ages 60-94. J Aging Phys Activity 1999;7(2):162-81 [\[LINK\]](#)
8. Gatenio-Hefling O, Tzemah-Shahar R, Asraf K, Dillian O, Gil E, Agmon M, Revisiting the "Timed Up and Go" test: a 12-s cut-off can predict Hospitalization Associated Functional Decline in older adults, GeroScience 47, 1039–1048 (2025) [\[LINK\]](#)
9. Bohannon RW. Reference values for the timed up and go test: a descriptive meta-analysis. J Geriatr Phys Ther. 2006;29: 64–8. [\[LINK\]](#)
10. Middleton A et al., Walking Speed: The Functional Vital Sign, J. of Aging and Physical Activity, 23:2, 2015 [\[LINK\]](#)
11. Quach L et al., The Non-linear Relationship between Gait Speed and Falls: The MOBILIZE Boston Study. J Am Geriatr Soc. 2011 Jun 7;59(6):1069–107 [\[LINK\]](#)
12. Jenkinson D, Mace M, Reliability of the Able Assess Falls Screening Tool for Assessing Mobility in Older Adults (manuscript in draft)



GRIPABLE FOR SCREENING FALLS RISK: INTERNAL REPORTS

Lead Institution	Title	Data/Evidence	Status
GRIPABLE	[1] The Multifactorial Evidence Base for Falls Risk Screening (Internal Report) <i>Able Assess</i>	The Able Assess Falls Risk Screening platform uses a multifactorial, evidence-based approach to identify individuals at risk of falling. The risk of falling is influenced by factors such as muscle strength, balance, mobility, and physical resilience. The Able Assess platform delivers a falls risk screening in under five minutes by combining four objective assessments.	Internal Report
	[2] Reliability of the Able Assess Falls Screening Tool for Assessing Mobility in Older Adults (Internal Report) <i>Daniel Jenkinson, Michael Mace</i>	CST: Good-to-Excellent inter-instrument agreement TUG: Excellent inter-instrument agreement 4GS: Moderate-to-Good inter-instrument agreement Indicative results demonstrating promising predictive capability as a meaningful predictor of falls history	Internal Report



GRIPABLE FOR GAMIFIED REHABILITATION

Rehabilitation plays a pivotal role in restoring and improving physical function and quality of life for individuals recovering from injuries or medical conditions. One key factor in achieving better outcomes is the intensity of rehabilitation exercises.

Higher Rehabilitation Repetition, Duration and Intensity for Better Outcomes:

1. **Enhanced Recovery:** Training with higher repetition and longer duration can lead to clinically meaningful improvements, whilst increased exercise intensity can promote muscle strength, endurance, and functional gains [1-3].
2. **Neuroplasticity:** Intensive rehabilitation can stimulate neuroplastic changes in the brain and nervous system, facilitating better recovery of motor skills and function [3].
3. **Patient Motivation and Gamification:** Engaging patients in higher-intensity rehabilitation can boost motivation and compliance with their treatment plans, ultimately contributing to better results. Gaming is an effective motivational tool shown to be more effective than conventional treatment [4].

GripAble as a Rehabilitation Tool:

1. **Higher Repetition and Duration:** GripAble's effectiveness in facilitating rehabilitation exercises has been demonstrated in several studies showcasing its potential to enhance patient outcomes, such as doubling exercise duration and eight times repetitions compared to standard care in stroke patients [5] (see Slide 13-17).
2. **Personalised Training Intensity:** GripAble offers the flexibility to tailor training intensity to individual users' needs, allowing for personalised exercises that accommodate both higher and lower grip strength levels.
3. **Engagement and Usability:** GripAble game-based rehabilitation exercises demonstrate 87% usability and 68% acceptability with stroke patients [6] and equally high in other neurological and musculoskeletal conditions (see Slide 14).

1. Crow J, Smith A. National Clinical Guideline for Stroke for the UK and Ireland: Part II-The challenges and opportunities posed for occupational therapists. British Journal of Occupational Therapy. 2023;03080226231188014 [\[LINK\]](#)
2. Lohse KR, Lang CE, Boyd LA. Is more better? Using metadata to explore dose-response relationships in stroke rehabilitation. Stroke. 2014 Jul;45(7):2053-8 [\[LINK\]](#)
3. Ward NS, Brander F, Kelly K. Intensive upper limb neurorehabilitation in chronic stroke: outcomes from the Queen Square programme. Journal of Neurology, Neurosurgery & Psychiatry. 2019 May 1;90(5):498-506 [\[LINK\]](#)
4. Doumas I, Everard G, Dehem S, Lejeune T. Serious games for upper limb rehabilitation after stroke: a meta-analysis. Journal of neuroengineering and rehabilitation. 2021 Dec;18:1-6 [\[LINK\]](#)
5. Broderick M, Almedom L, Burdet E, Burrige J, Bentley P. Self-directed exergaming for stroke upper limb impairment increases exercise dose compared to standard care. Neurorehabilitation and Neural Repair. 2021 Nov;35(11):974-85. [\[LINK\]](#)
6. Broderick M, O'Shea R, Burrige J, Demain S, Johnson L, Bentley P. Examining Usability, Acceptability, and Adoption of a Self-Directed, Technology-Based Intervention for Upper Limb Rehabilitation After Stroke: Cohort Study. JMIR Rehabilitation and Assistive Technologies. 2023 Aug 21;10:e45993. [\[LINK\]](#)

Lead Institution	Title	Phase	Condition	Data/Evidence	Status
IMPERIAL	[1] Democratizing Neurorehabilitation: How Accessible are Low-Cost Mobile-Gaming Technologies for Self-Rehabilitation of Arm Disability in Stroke? (2016) [LINK] <i>Paul Rinne, Michael Mace, Tagore Nakornchai, Karl Zimmerman, Susannah Fayer, Pankaj Sharma, Jean-Luc Liardon, Etienne Burdet, Paul Bentley</i>	Phase 0 - Proof of concept	Stroke	Hand grip enabled 93% usability with impaired stroke patients	Completed and published
	[2] Balancing the playing field: collaborative gaming for physical training (2017) [LINK] <i>Michael Mace, Nawal Kinany, Paul Rinne, Anthony Rayner, Paul Bentley and Etienne Burdet</i>	Phase 0 - Proof of concept	Stroke / healthy	Two-player exergaming enhanced less-abled players' performance	
	[3] Elasticity improves handgrip performance and user experience during visuomotor control (2017) [LINK] <i>Michael Mace, Paul Rinne, Jean-Luc Liardon, Catherine Uhomiohi, Paul Bentley and Etienne Burdet</i>	Phase 0 - Proof of concept	Stroke / healthy	3x more users prefer elastic compared to rigid handgrip	
	[4] Optimizing self-exercise scheduling in motor stroke using Challenge Point Framework Theory (2019) [LINK] <i>Rajinder Lotay, Mike Mace, Paul Rinne, Etienne Burdet, Paul Bentley</i>	Phase 0 - Proof of concept	Stroke	Adaptive training task variability and difficulty yielded greater long-term performance	

GRIPABLE FOR REHABILITATION: PUBLICATIONS AND PROJECTS

Lead Institution	Title	Phase	Condition	Data/Evidence	Status
 Imperial College Healthcare NHS Trust	[5] Self-Directed Exergaming for Stroke Upper Limb Impairment Increases Exercise Dose Compared to Standard Care (2021) [LINK] <i>Michelle Broderick, Leeza Almedom, Etienne Burdet, Jane Burrridge, Paul Bentley</i>	Phase 1 - Feasibility	Stroke	Increased exercise duration 2-fold, and repetitions 8-fold, compared to standard care	Published
	[6] Examining Usability, Acceptability, and Adoption of a Self-Directed, Technology-Based Intervention for Upper Limb Rehabilitation After Stroke: Cohort Study (2023) [LINK] <i>Michelle Broderick, Robert O'Shea, Jane Burrridge, Sara Demain, Louise Johnson, Paul Bentley</i>	Phase 1 - Feasibility	Stroke	The technology was usable and acceptable for most participants, with adoption linked to usability, acceptability, and stroke severity.	
	[7] Multicentre pilot randomised control trial of a self-directed exergaming intervention for poststroke upper limb rehabilitation: research protocol (2024) [LINK] <i>Michelle Broderick, Jane Burrridge, Sara Demain, Louise Johnson, Robert O'Shea, Paul Bentley</i>	Phase 2 - Protocol	Stroke	Protocol only	Published
	[8] Self-directed exergaming for upper Limb rehabilitation post-stroke – a multicentre pilot randomised control trial [Conference Presentation] (2024); UK Stroke Forum, Birmingham, UK <i>Michelle Broderick, Joe Brereton, Dr Robert O'Shea, Dr Louise Johnson PhD, Prof Jane Burrridge, Prof Sara Demain, Dr Paul Bentley</i>	Phase 2	Stroke	Over 10 hours of GripAble training led to significant gains in arm function (7 points on the Fugl-Meyer scale) and hand dexterity (12 points on the Action Research Arm Test).	Accepted for presentation - Full manuscript in draft
	[9] Leaving stroke survivors to their own devices: A qualitative study of stroke survivors' experiences with a technology-facilitated, self-directed intervention early after stroke [Conference Presentation] (2024) <i>Michelle Broderick, Dr Paul Bentley</i>				

G GRIPABLE FOR REHABILITATION: PUBLICATIONS AND PROJECTS

Lead Institution	Title	Phase	Conditions	Data/Evidence	Status
 luzerner kantonsspital LUZERN SURSEE WOLHUSEN	[10] Combining repetitive transcranial magnetic stimulation and video game-based training to improve dexterity in Parkinson's disease: study protocol of a randomized controlled trial (2021) [LINK] <i>Manuela Pastore-Wapp, Dirk Lehnick, Tobias Nef, Stephan Bohlhalter, Tim Vanbellinghen</i>	Phase 2 - Protocol	Parkinson's disease	Study protocol for determining the effectiveness of video based therapy for PD	Completed and published
	[11] Usability of Two New Interactive Game Sensor-Based Hand Training Devices in Parkinson's Disease (2022) [LINK] <i>Lea Saric, Samuel E. J. Knobel, Manuela Pastore-Wapp, Tobias Nef, Fred W. Mast, Tim Vanbellinghen</i>	Phase 1 - Feasibility	Parkinson's disease	High usability of GripAble for persons with PD and healthy participants	
	[12] Feasibility of a combined intermittent theta-burst stimulation and video game-based dexterity training in Parkinson's disease (2023) [LINK] <i>Manuela Pastore-Wapp, Brigitte C. Kaufmann, Thomas Nyffeler, Simona Wapp, Stephan Bohlhalter, Tim Vanbellinghen</i>	Phase 1 - Feasibility	Parkinson's disease	Good usability reported by patients	
 UNIVERSITY OF OXFORD	[13] Improving sleep and learning in rehabilitation after stroke, part 2 (INSPIRES2): study protocol for a home-based randomised control trial of digital cognitive behavioural therapy (dCBT) for insomnia (2023) [LINK] <i>Matthew Weightman, Barbara Robinson, Ricky Fallows, Alasdair L Henry, Simon D Kyle, Emma Garratt, Anton Pick, Rachel Teal, Sara Ajina, Nele Demeyere, Colin A Espie, Ben Seymour, Heidi Johansen-Berg, Melanie K Fleming</i>	Phase 3 - Protocol	Stroke	Protocol published – study ongoing	Protocol published and study ongoing

G GRIPABLE FOR REHABILITATION: PUBLICATIONS AND PROJECTS

Lead Institution	Title	Phase	Condition s	Data/Evidence	Status
 Centro Hospitalar Universitário do Algarve	[14] Effectiveness on the use of the GripAble platform in distal radius fracture: case study (2022) [LINK] <i>Carlota Vieira, Catarina Antão, António Arsénio Duarte, Ana Paula Martins, José Luís Coelho</i>	Phase 0 - Case study	Distal radius fracture	GripAble significantly improves pain, joint range and grip strength	Completed and published
 IPSEN Innovation for patient care GRIPABLE	[15] Single-Center, Prospective, Observational Study to Assess the Use of the Rehabilitation Tool, GripAble, in Adults With Upper Limb Spasticity Receiving Botulinum Neurotoxin Type A in the UK(2023) [LINK] <i>Abayomi Salawu, Sharon Scott, Sharah Abdul Mutalib, Vadim Degtiar, Françoise Calvi-Gries, Mario Ippolito, Mounika Parimi, Stephanie Castello, Tony Cosgrove</i>	Phase 1 - Feasibility	Stroke	Results show indication of potential benefit of combining GripAble with BoNT-A treatment. GripAble allowed remote monitoring of patients between BoNT-A cycles	Completed and published
 Pace	[16] To evaluate the effects of GripAble on hand function in children with cerebral palsy (2022) <i>Rachel Murill, Roshni Khatri</i>	Phase 1 - Feasibility	Cerebral palsy	GripAble improves grip and pinch strength, as well as a self reported improvement in ADLs.	Article in press

G GRIPABLE FOR REHABILITATION: PUBLICATIONS AND PROJECTS

Lead Institution	Title	Phase	Conditions	Data/Evidence	Status
	[17] An intensive exercise program using a technology-enriched rehabilitation gym for the recovery of function in people with chronic stroke: usability study (2023) [LINK] <i>Andy Kerr, Maisie Keogh, Milena Slachetka, Madeleine Grealy, Philip Rowe</i>	Phase 1 - Feasibility	Stroke	GripAble is highly usable and recommended for home-based rehabilitation alongside other small-tech	Published and ongoing
	[18] A co-creation centre for accessible rehabilitation technology (2022) [LINK] <i>Andy Kerr, Madeleine Ann Grealy, Anja Kuschmann, Rosie Rutherford, Philip Rowe</i>	Exploratory study	Stroke	Supports exploration of a personalised, self-managed approach with technology as a key enabler	
	[19] Flexor Injury Rehabilitation Splint Trial (FIRST): protocol for a pragmatic randomised controlled trial comparing three splints for finger flexor tendon repairs (2024) <i>Emma Bamford, Hannah Berntsson, Suzanne Beale, Lauren Desoysa, Joseph Dias, Sienna Hamer-Kiwacz, Daniel Hind, Nick Johnson, Amanda Loban, Kaye Molloy, Emma Morvan, Ines Rombach, Anna Selby, Praveen Thokala, Chris Turtle, Stephen Walters & Avril Drummond</i>	Phase 3 - Protocol	Finger flexor tendon repair	Protocol only - study ongoing	Protocol published
	[20] Enhancing self-directed arm exercise practice using GripAble gaming device and Lycra arm sleeve in people with stroke: An Evaluative Study (2022) <i>Praveen Kumar, Alex Pearce</i>	Phase 1 - Feasibility	Stroke	Pending full data analysis / report	Completed pending publication

CHANGE HISTORY

Version	Date of revision	Descriptions
V1	September 2023	Initial version of GripAble's supporting evidence
V2	March 2024	Included evidence pertaining to grip strength and its correlation with cancer, cardiovascular disease, Alzheimer's disease, diabetes, and obesity.
V3	August 2024	Added new publications
V4	January 2026	Added GripAble for Screening Falls Risk section and updated publication